



Lecture02

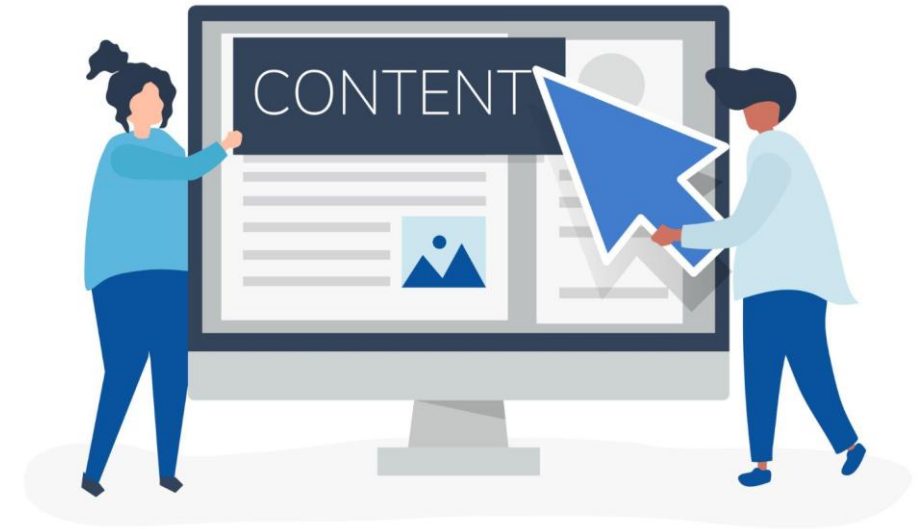
Languages and Operations on Languages

Automata Theory

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1. More on Alphabets and Strings
2. Languages
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1. MORE ON ALPHABETS AND STRINGS

1.1 Power of an Alphabet

- ❖ We use exponential notation to express the set of all strings of a certain length from an alphabet
- ❖ We define to be the set of strings of length each of whose symbols is in
- ❖ Some example: X^k to be the set of strings of length k each of whose symbols is in X
 - $X^2 = \{aa, ab, ba, bb\}$
 - $X^3 = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$
- ❖ Note that $X^0 = \{\varepsilon\}$ regardless of alphabet X
- ❖ Difference between X and X^1
 - X is an alphabet (its members are symbols)
 - X^1 is set strings (its members are strings of length 1)

1. MORE ON ALPHABETS AND STRINGS

1.2 All Strings Over an Alphabet

- ❖ The set of all strings over an alphabet X is denoted X^*
 - Example: $\{a, b\}^* = \{\varepsilon, a, b, aa, ab, \dots\}$
 - Put another way $X^* = X^0 \cup X^1 \cup X^2 \cup \dots$
- ❖ X^+ is used if we want to exclude the empty string ε from the set of string
 - $X^+ = X^1 \cup X^2 \cup \dots$

1. MORE ON ALPHABETS AND STRINGS

1.3 Concatenation of Strings

- ❖ Let x and y be strings then. The concatenation of two strings x and y is a new string from by making a copy of x and following it by a copy of y
- ❖ $x \cdot y$ (or simply xy) denotes the concatenation of x and y
- ❖ If x is a string composed of i symbols $x = a_1 a_2 \dots a_i$ and y composed of j symbols $y = b_1 b_2 \dots b_j$ the xy is the string of length $i + j$: $xy = a_1 a_2 \dots a_i b_1 b_2 \dots b_j$
 - ✓ **Example:** $x = abbab$ and $y = aab$ then $xy = abbabaab$
- ❖ Some remarks:
 - $xy \neq yx$
 - $(xy)z = x(yz)$
 - $x\varepsilon = \varepsilon x = x$

1. MORE ON ALPHABETS AND STRINGS

1.3 Power of a String

- ❖ We can also use exponential notation on strings
- ❖ A power k of a string x , denoted x^k is defined by:
$$\begin{cases} x^0 = \varepsilon \\ x^k = x^{k-1}x \quad \forall k \geq 1 \end{cases}$$
- ❖ Example: Let $x = ab$ be a string, then $x^2 = abab$, $x^3 = ababab$
- ❖ Some remarks:
 - $|x^i| = i|x|$
 - $\forall i, j \geq 0, x^i y^j = x^j x^i = x^{i+j}$

2. LANGUAGES

2.1 Definition

- ❖ A set of strings all of which are chosen from X^* , where X is an alphabet, is called a language
- ❖ If X is an alphabet and $L \subseteq X^*$, then L is a language over X
- ❖ A language can be finite or infinite
- ❖ Common languages can be viewed as sets of strings. An example is English language where collected of legal English word is a set of string over the alphabet $\{a, b, c, \dots, z\}$

2. LANGUAGES

2.1 Definition (cont.)

- ❖ There are also many other languages that appear in this course. Some abstract examples are:
 1. The language which consists of four strings $\{ab, ba, aab, bba\}$
 2. The language of all strings consisting of n a 's followed by n b 's, for $n \geq 0$: $\{\varepsilon, ab, aabb, aaabbb\}$
 3. The set of strings of a 's and b 's with equal number of each: $\{\varepsilon, ab, ba, aabb, abab, baab, \dots\}$
 4. The set of binary numbers whose value is a prime: $\{10, 11, 101, 111, 1011, \dots\}$
 5. X^* is a language consisting of all strings over an alphabet X
 6. $\emptyset$, the empty language, contains no string. It is a language over any alphabet
 7. $\{\varepsilon\}$, the language consisting of only the empty string, is also a language over any alphabet
- ❖ Notice that $\emptyset \neq \{\varepsilon\}$

2. LANGUAGES

2.1 Set-Former

- ❖ It is common to describe a language using a set-former: $\{w \mid \text{something about } w\}$
- ❖ The expression above is read “the set of word such that ...”
- ❖ Some examples:
 - $\{w \mid w \text{ consists of an equal number of } a\text{'s and } b\text{'s}\}$
 - $\{w \mid w \text{ is a syntactically correct } C \text{ program}\}$
- ❖ It is also common to replace w by some expression with parameters
 - $\{a^n b^n \mid n \geq 1\}$
 - $\{a^i b^j \mid 0 \leq i \leq j\}$

3. OPERATIONS ON LANGUAGES

3.1 Union and Product

❖ A union between two language L_1 and L_2 : $L_1 \cup L_2 = \{w \in L_1 \text{ or } w \in L_2\}$

✓ **Example:** $L_1 = \{ab, bba\}$, $L_2 = \{bba, ba, baa\} \Rightarrow L_1 \cup L_2 = \{ab, bba, ba, baa\}$

❖ A product (or concatenation) of two languages L_1 and L_2 :

$$L_1 \cdot L_2 = \{w_1 w_2 \text{ such that } w_1 \in L_1 \text{ and } w_2 \in L_2\}$$

❖ Noted that $L_1 \cdot L_2 \neq L_2 \cdot L_1$

✓ **Example:**

➤ $L_1 = \{a, ab\}$, $L_2 = \{b, ba\} \Rightarrow L_1 \cdot L_2 = \{ab, aba, abb, abbba\}$

➤ Concatenating with the empty set L . $\emptyset = \emptyset \cdot L = \emptyset$

3. OPERATIONS ON LANGUAGES

3.2 Closure

- ❖ A power k of a language L , denoted L^k is defined by:

$$\begin{cases} L^0 = \{\varepsilon\} \\ L^k = L^{k-1} \cdot L \quad \forall k \geq 1 \end{cases}$$

✓ **Example:** $L = \{a, ab\} \Rightarrow L^2 = L \cdot L = \{aa, aab, aba, abab\}$

- ❖ A Kleene star (or Kleene closure) of a language L : $L^* = \bigcup_{i=0}^{\infty} L^i$. L^* is the language obtained from concatenating a number of times of strings in L (possibly zero times).

✓ **Example:** $L = \{a, ab\} \Rightarrow L^* = \{\varepsilon, a, ab, aa, aab, \dots\}$

- ❖ A positive closure of a language L : $L^+ = \bigcup_{i=0}^{\infty} L^i$

➤ $L^+ \subseteq L \quad \forall L$

➤ $L^* = L^+ \cup L^0$

Question: $L^+ = L^* \setminus \{\varepsilon\}$?

3. OPERATIONS ON LANGUAGES

3.2 Closure (cont.)

❖ Some remarks:

1. $L^+ = L^* \setminus \{\varepsilon\}$ if and only if $\varepsilon \notin L$
2. $L^+ = L.L^* = L^*.L$
3. $\{\varepsilon\}^* = \{\varepsilon\}^+ = \{\varepsilon\}$
4. $L.L^+ = L^+.L$
5. $(L^+)^+ = L^+$
6. $(L^+)^* = (L^*)^+ = (L^*)^* = L^*$
7. $L_1 \subset L_2 \Rightarrow L_1^+ \subset L_2^+$ and $L_1^* \subset L_2^*$

3. OPERATIONS ON LANGUAGES

3.3 Mirror of Languages

❖ A mirror of a language L : $L^R = \{W^R, W \in L\}$

✓ **Example:** $L = \{ab, abb\} \Rightarrow L^R = \{ba, bba\}$

❖ Some remake:

1. $(L_1 \cup L_2)^R = L_1^R \cup L_2^R = \{W^R \mid w \in L_1 \text{ or } w \in L_2\}$

2. $(L_1 \cdot L_2)^R = L_2^R \cdot L_1^R$

3. $(L^R)^R = L$

4. $(L^+)^R = (L^R)^+$

5. $(L^*)^R = (L^R)^*$

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